

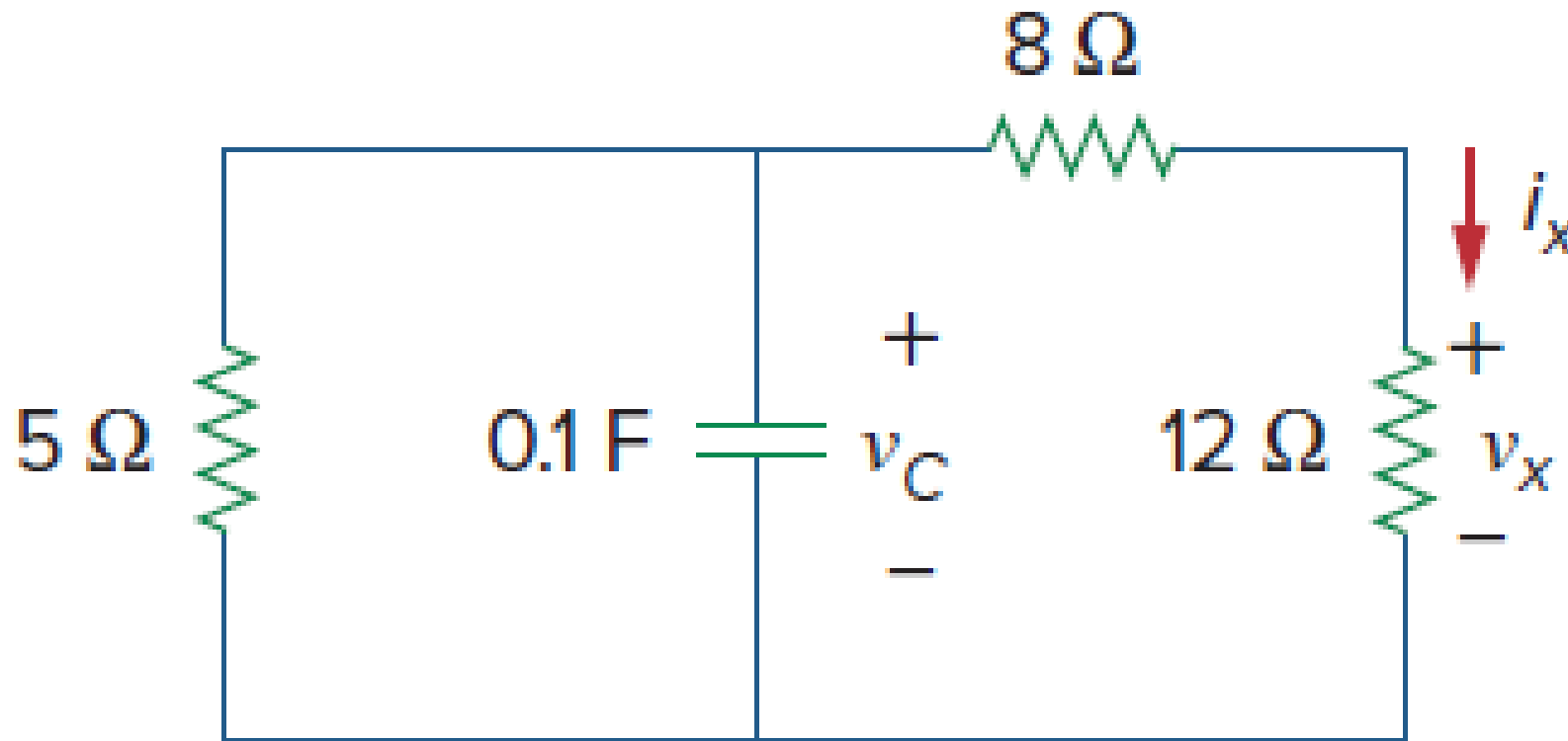
SOLVING NUMERICALS

Points to remember :

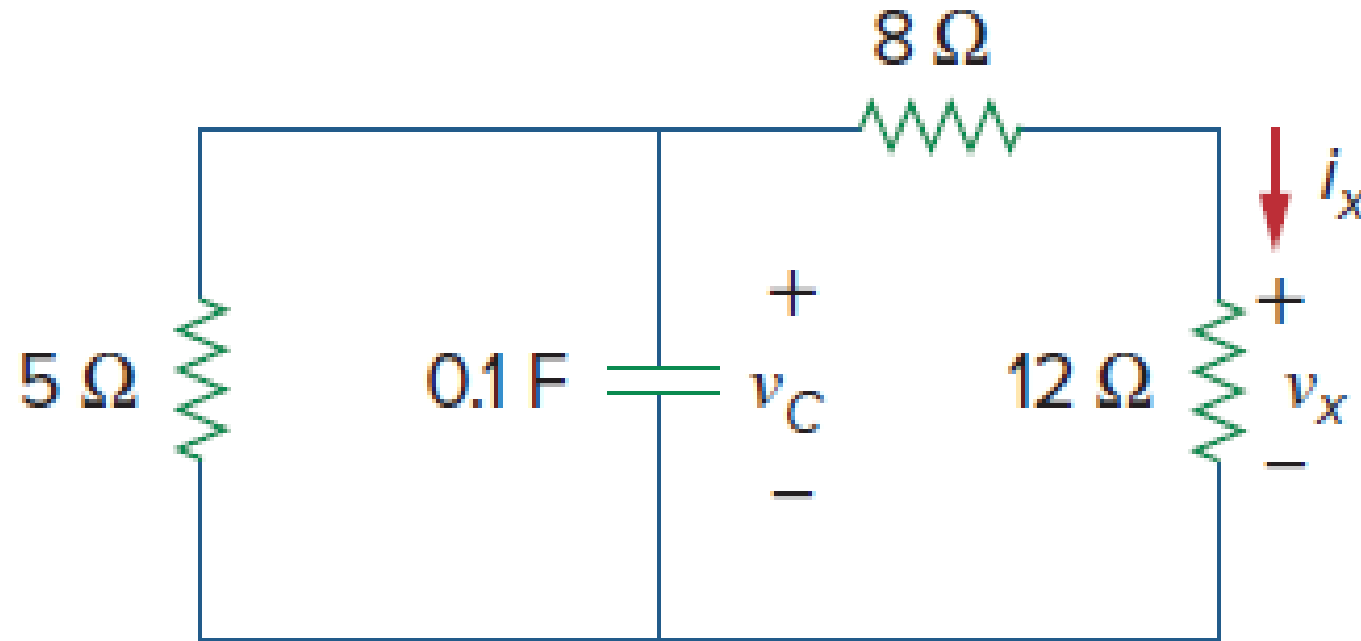
Elements	DC steady state	Continuous quantity (from $t=0^-$ to $t=0^+$)
R	R	-
L	Short-circuit ($v = 0$)	Current i
C	Open-circuit ($i = 0$)	Voltage v

SOURCE-FREE RC CIRCUIT

Q1. Consider the circuit below. Let $v_C(0)=15$ V. Find v_C , v_x and i_x for $t > 0$.



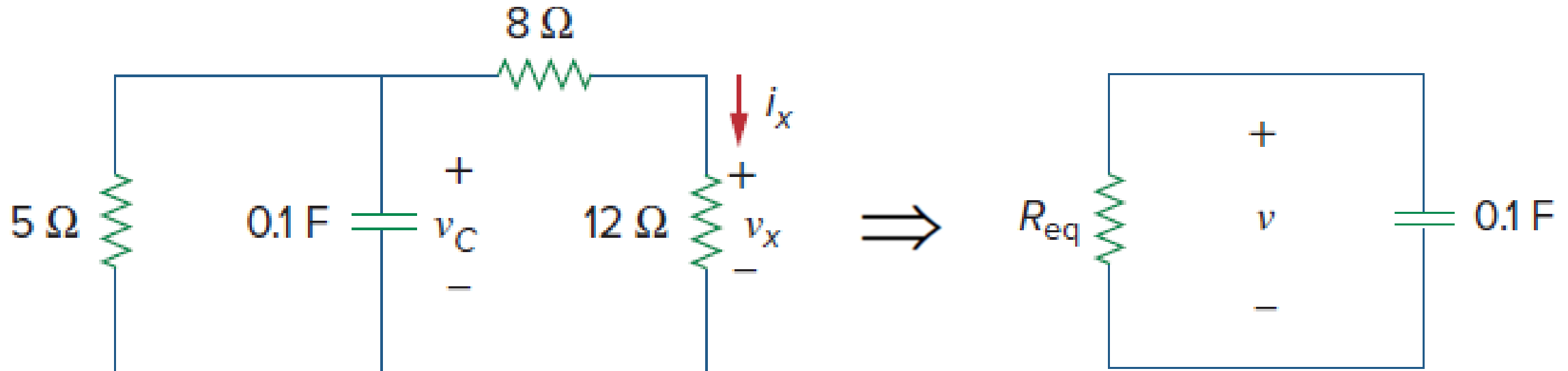
SOURCE-FREE RC CIRCUIT



Solution :

- We first convert the given circuit into a simple R - C circuit.
- Find the equivalent resistance or the Thevenin resistance at the capacitor terminals.

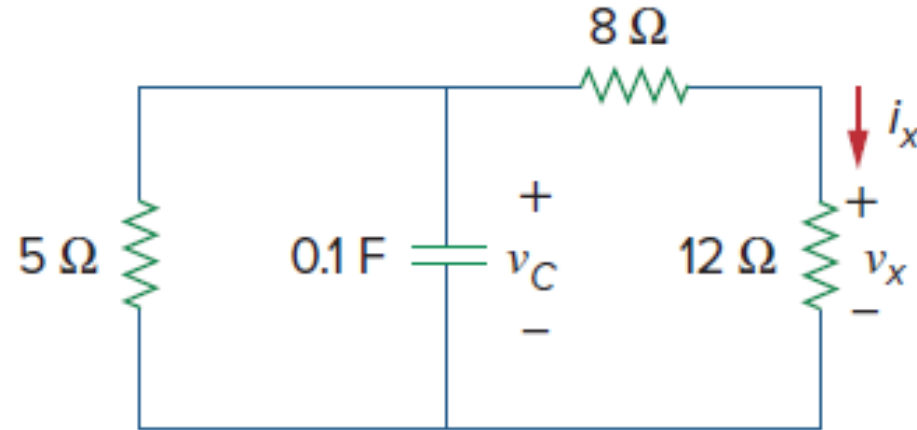
SOURCE-FREE RC CIRCUIT



$$R_{eq} = \frac{20 \times 5}{20 + 5} = 4\ \Omega$$

The time constant is $\tau = R_{eq} C = 4(0.1) = 0.4\text{ s}$

SOURCE-FREE RC CIRCUIT



$$v(t) = V_0 e^{-t/\tau} = 15 e^{-t/0.4} \text{ V} = 15 e^{-2.5t} \text{ V}$$

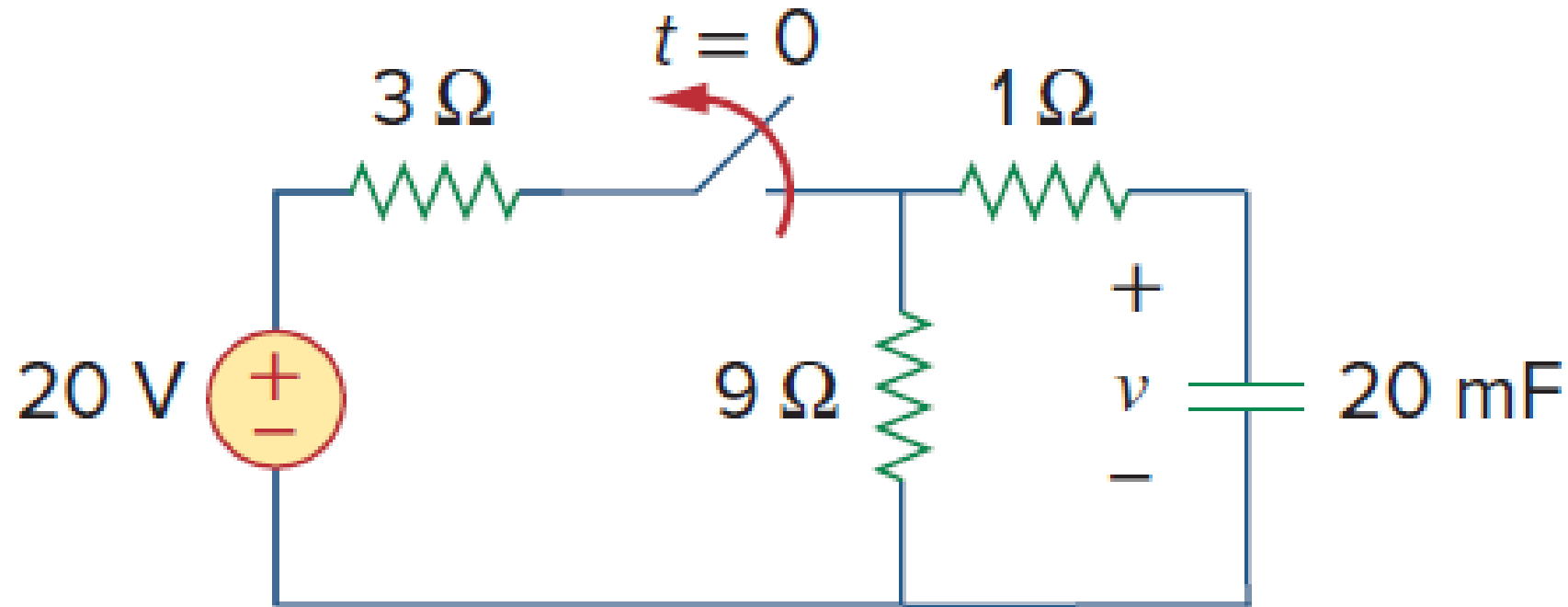
we can use voltage division to get v_x

$$v_x = \frac{12}{12 + 8} v_C = 0.6(15e^{-2.5t}) = 9e^{-2.5t} \text{ V}$$

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t} \text{ A}$$

SOURCE-FREE RC CIRCUIT

Q2. The switch in the circuit below is closed for a long time, and then opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Also calculate the energy stored in the capacitor before opening of the switch.



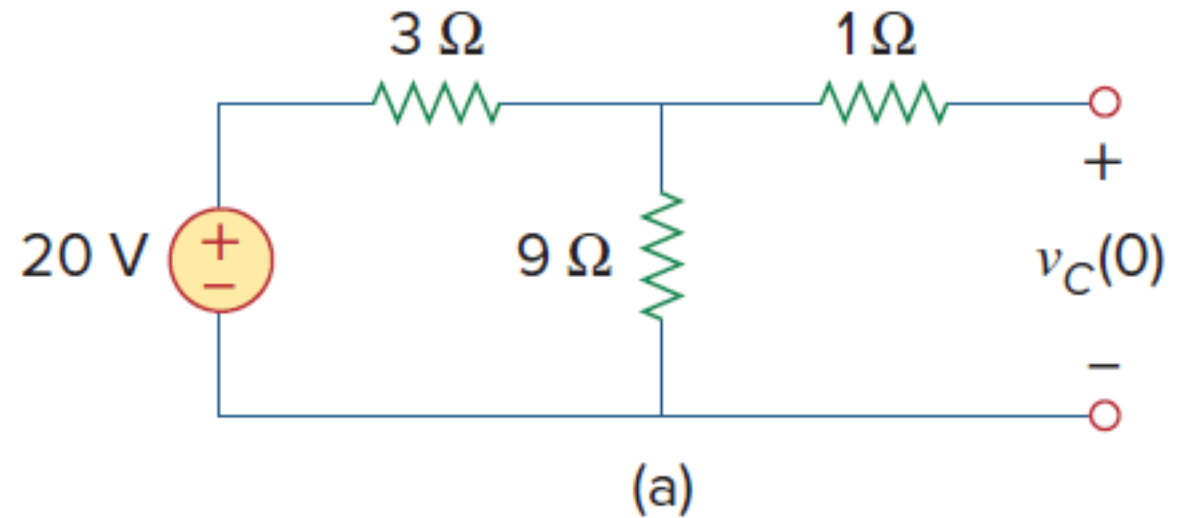
SOURCE-FREE RC CIRCUIT

Solution:

For $t < 0$, the switch is closed and the capacitor is an open circuit in steady state, as represented in Fig.(a).

Using voltage division

$$v_C(t) = \frac{9}{9+3}(20) = 15 \text{ V}, \quad t < 0$$



Since the voltage across a capacitor cannot change instantaneously, the voltage across the capacitor at $t = 0^-$ is the same at $t = 0^+$, or

$$v_C(0) = V_0 = 15 \text{ V}$$

SOURCE-FREE RC CIRCUIT

Solution:

For $t > 0$, the switch is open, and we have the RC circuit shown in Fig. (b),

$$R_{eq} = 1 + 9 = 10\Omega$$

The time constant is

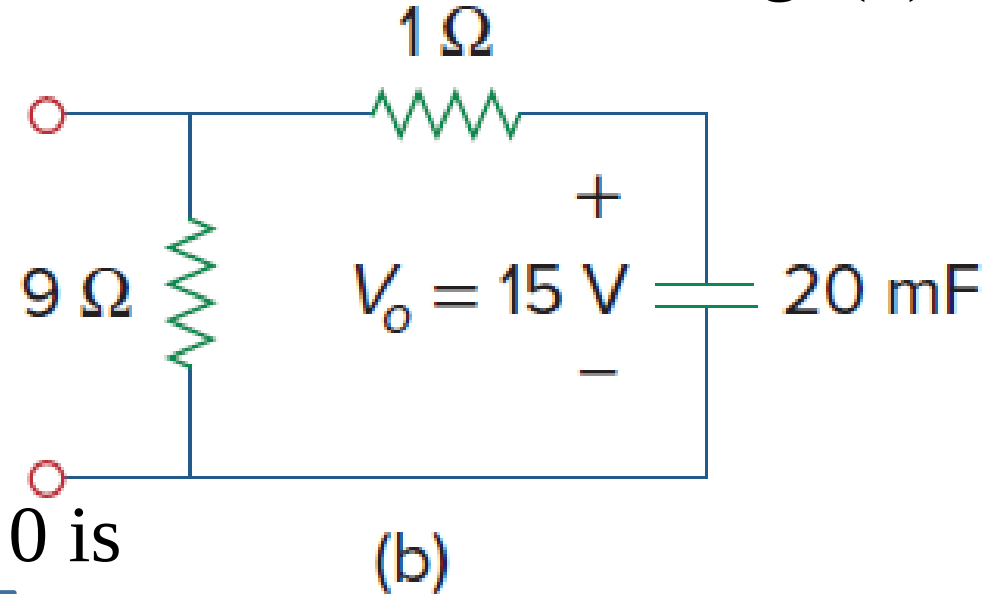
$$\tau = R_{eq} C = 10 \times 20 \times 10^{-3} = 0.2\text{ s}$$

Thus, the voltage across the capacitor for $t \geq 0$ is

$$v(t) = v_C(0) e^{-t/\tau} = 15 e^{-t/0.2} \text{ V} = 15 e^{-5t} \text{ V}$$

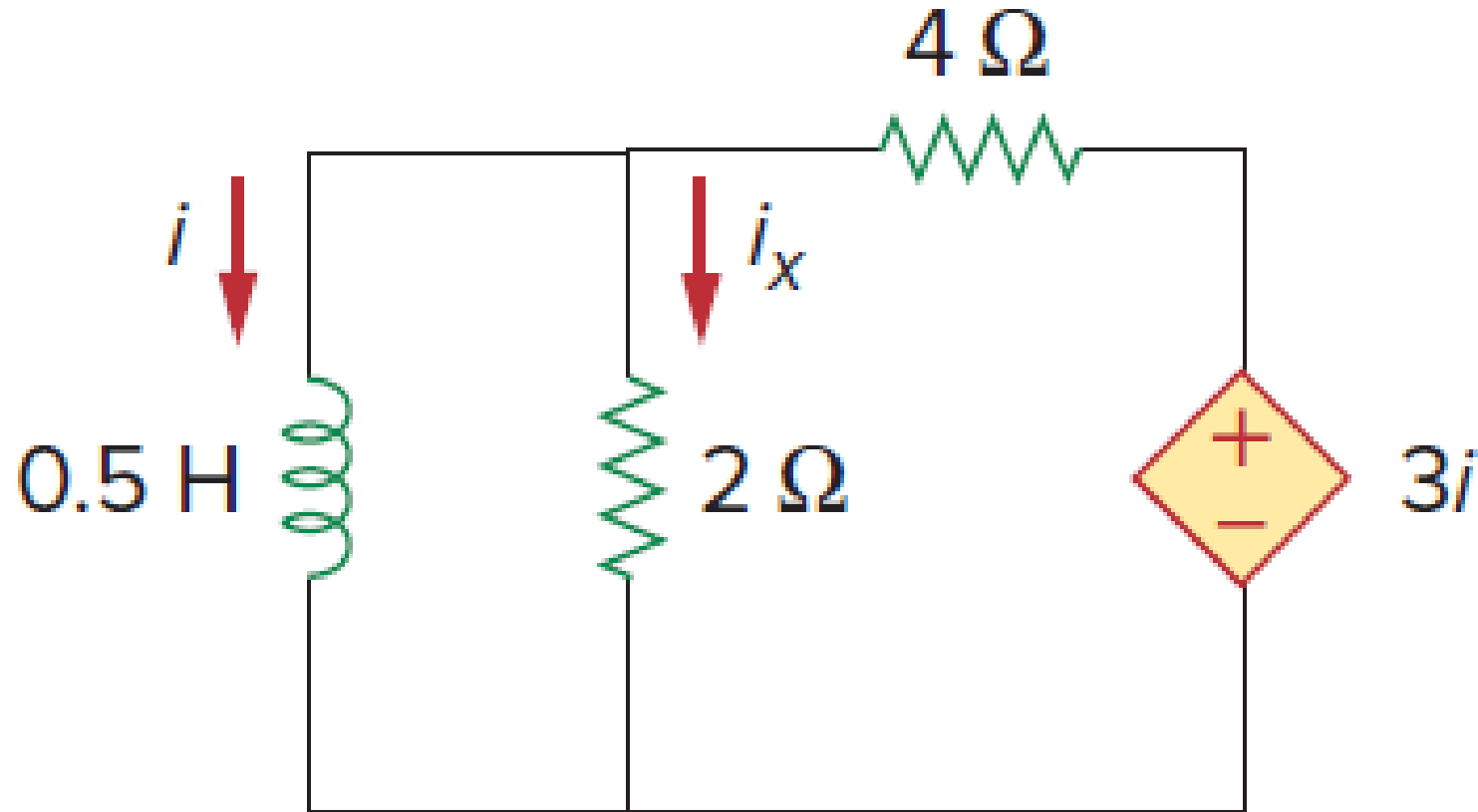
The initial energy stored in the capacitor is:

$$w_C(0) = \frac{1}{2} C v_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \text{ J}$$



SOURCE-FREE RL CIRCUIT

Q3. Assuming that $i(0) = 10$ A, calculate $i(t)$ and $i_x(t)$ in the circuit below



SOURCE-FREE RL CIRCUIT

Method-2:

Applying KVL to the circuit For loop 1,

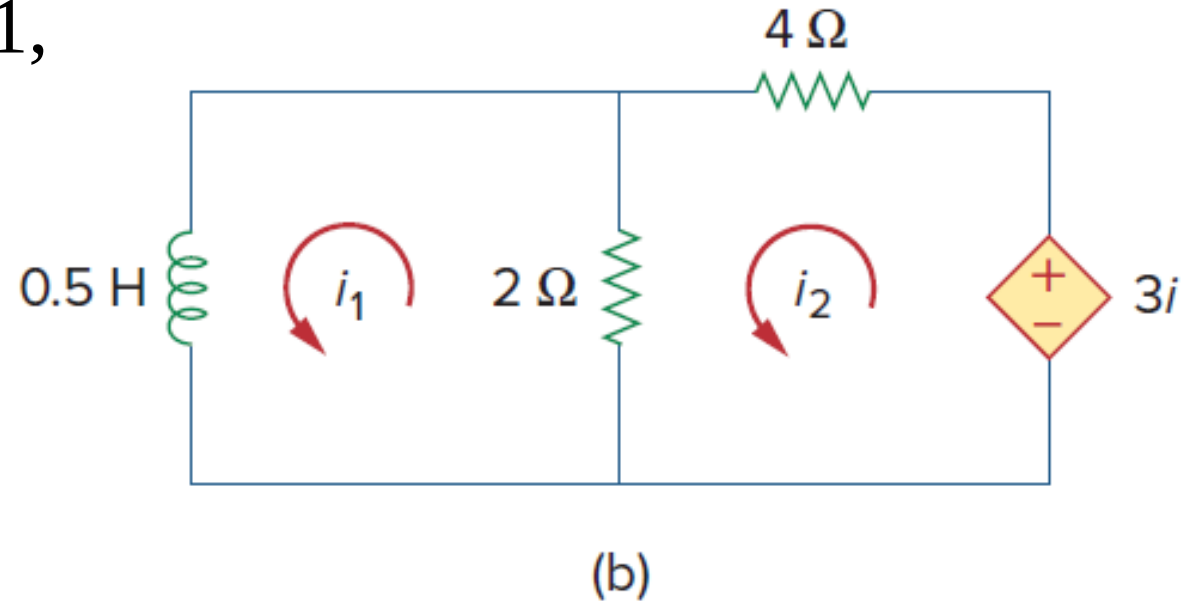
$$\frac{1}{2} \frac{di_1}{dt} + 2(i_1 - i_2) = 0 \quad (3)$$

For loop 2,

$$6i_2 - 2i_1 - 3i_1 = 0 \Rightarrow i_2 = \frac{5}{6} i_1$$

Substituting above into Eq. (3) gives $\frac{di_1}{dt} + \frac{2}{3} i_1 = 0$

$$\Rightarrow \ln \frac{i(t)}{i(0)} = -\frac{2}{3} t \Big|_0^t \Rightarrow i(t) = i(0)e^{-(2/3)t} = 10e^{-(2/3)t} \text{ A}, \quad t > 0$$



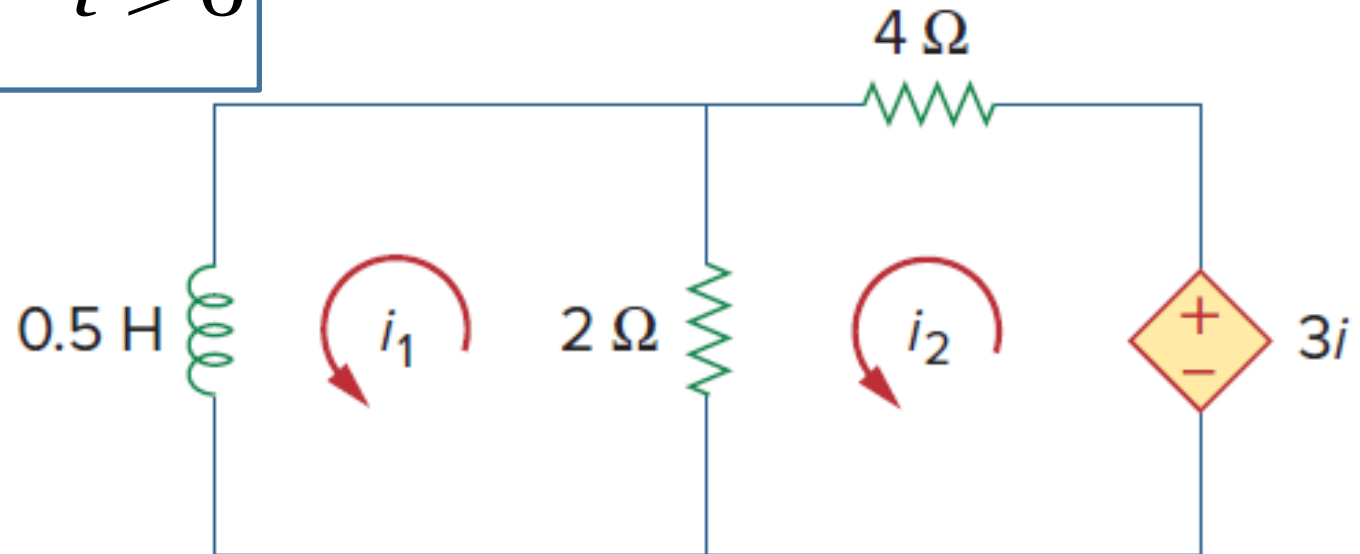
SOURCE-FREE RL CIRCUIT

The voltage across the inductor is

$$v = L \frac{di}{dt} = 0.5(10) \left(-\frac{2}{3} \right) e^{-(2/3)t} = -\frac{10}{3} e^{-(2/3)t} \text{ V}$$

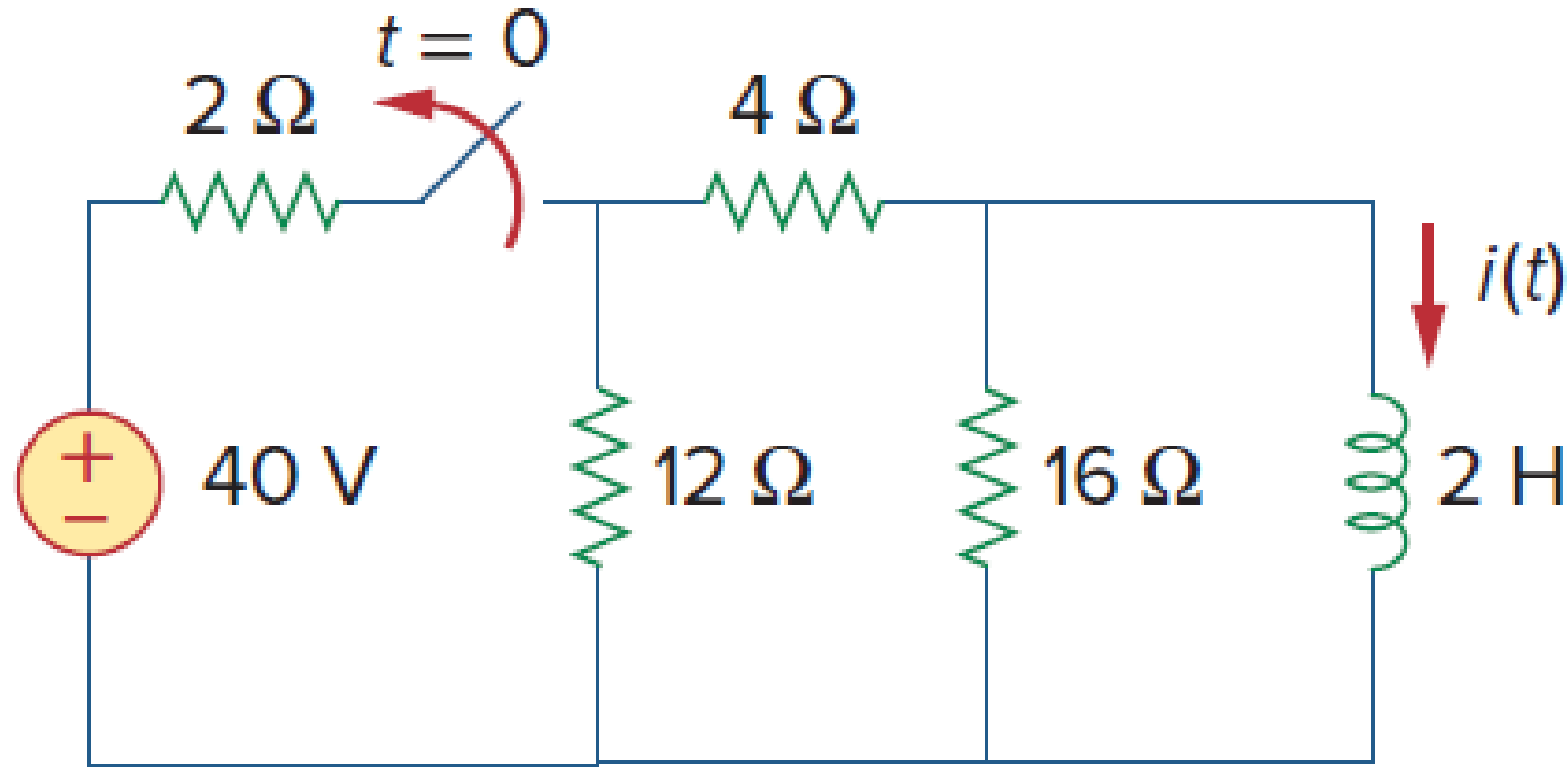
Since the inductor and the 2- Ω resistor are in parallel,

$$i_x(t) = \frac{v}{2} = -1.6667 e^{-(2/3)t} \text{ A}, \quad t > 0$$



SOURCE-FREE RL CIRCUIT

Q4. The switch in the circuit below is closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.



SOURCE-FREE RL CIRCUIT

Solution:

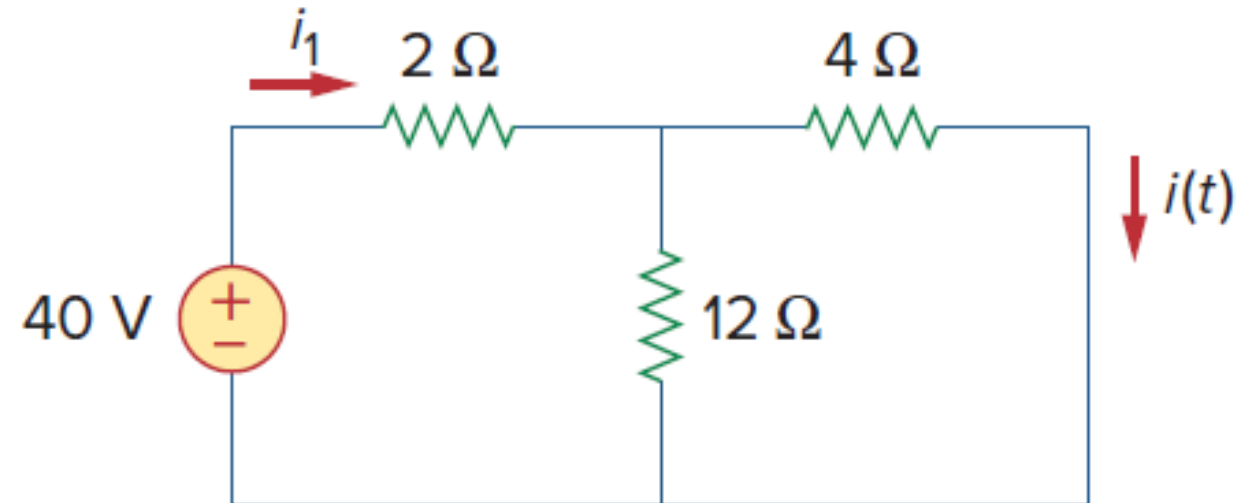
For $t < 0$, the switch is closed, and the inductor acts as a short circuit in steady state. The $16\text{-}\Omega$ resistor is short-circuited; the resulting circuit is shown in Fig (a).

$$R_{eq} = \frac{4 \times 12}{4 + 12} + 2 = 5\Omega$$

$$i_1 = \frac{40}{5} = 8\text{ A}$$

We obtain $i(t)$ from i_1 using current division,

$$i(t) = \frac{12}{12 + 4} i_1 = 6\text{ A}, t < 0$$



(a)

SOURCE-FREE RL CIRCUIT

Since the current through an inductor cannot change instantaneously,

$$i(0) = i(0^-) = 6A$$

For $t > 0$, the switch is open and the voltage source is disconnected.

We now have the source-free RL circuit in Fig.(b).

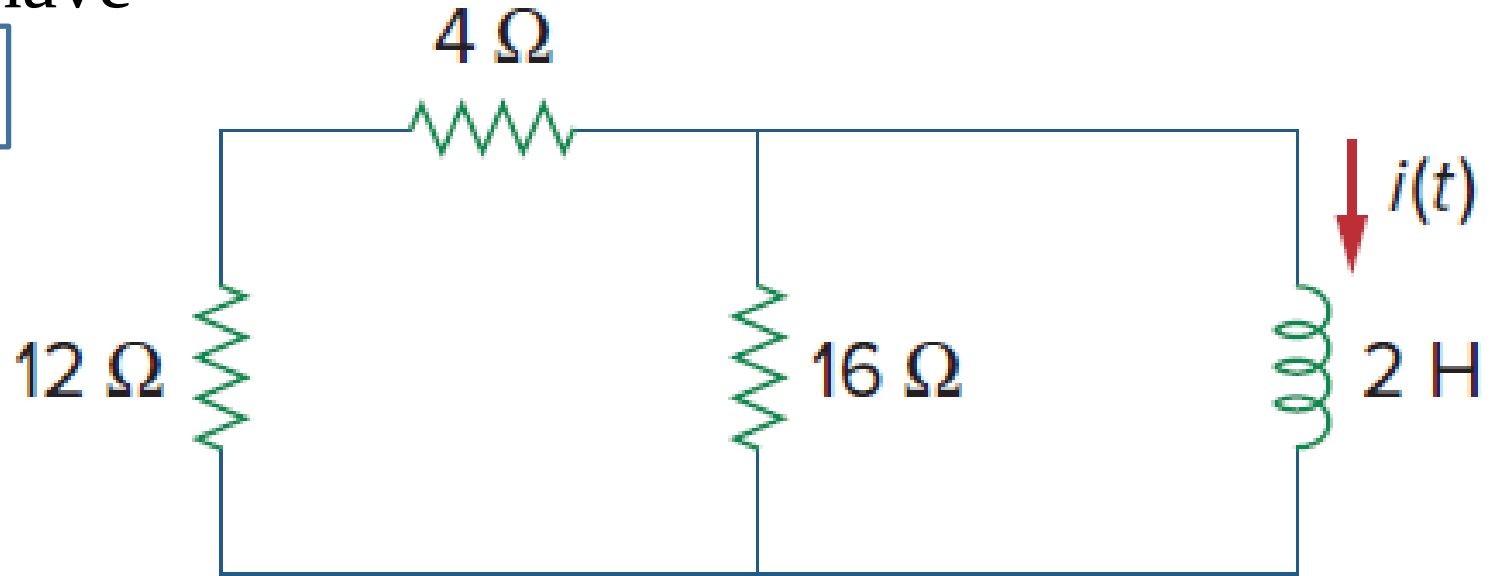
Combining the resistors, we have

$$R_{eq} = (12 + 4) \parallel 16 = 8 \Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = \frac{1}{4} s$$

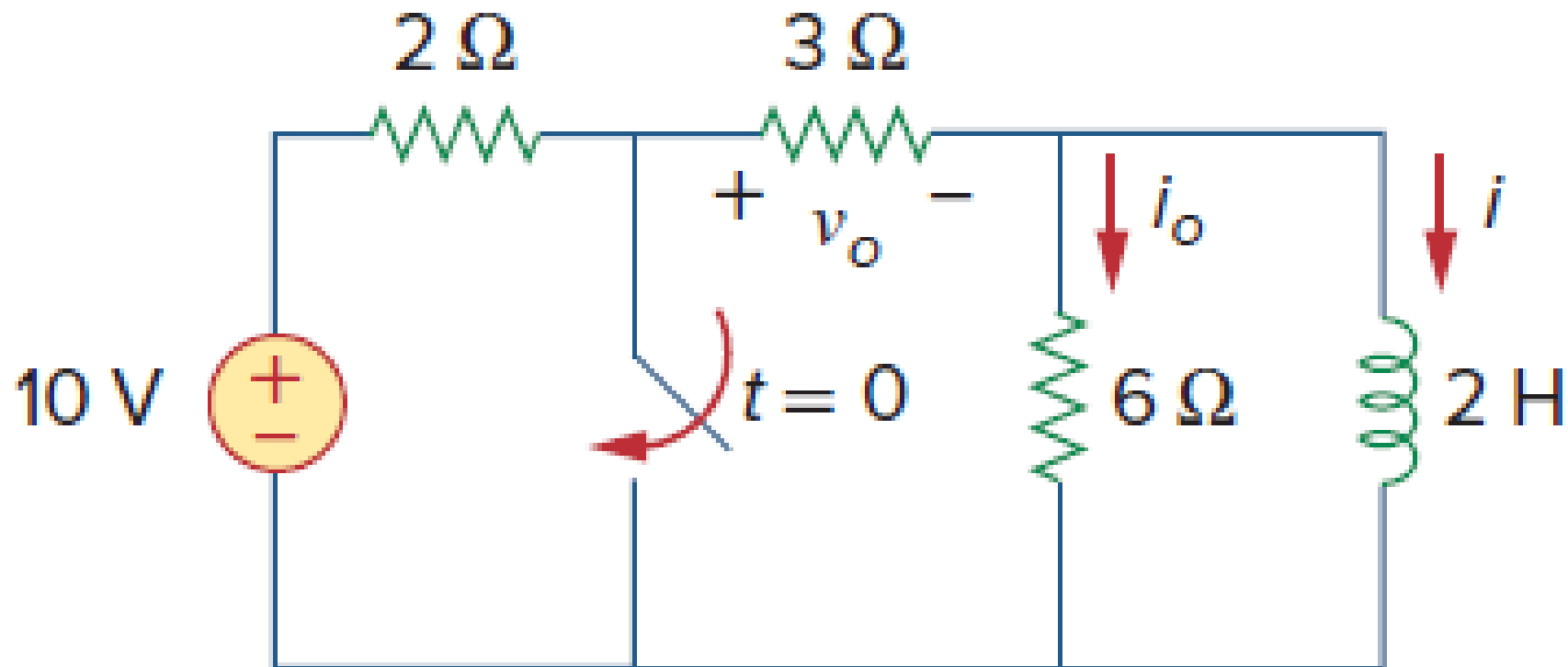
$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t} A$$



(b)

SOURCE-FREE RL CIRCUIT

Q5. In the circuit shown below, find i_o , v_o , and i for all $t > 0$, assuming that the switch was open for a long time and closed at $t = 0$.



SOURCE-FREE RL CIRCUIT

Solution :

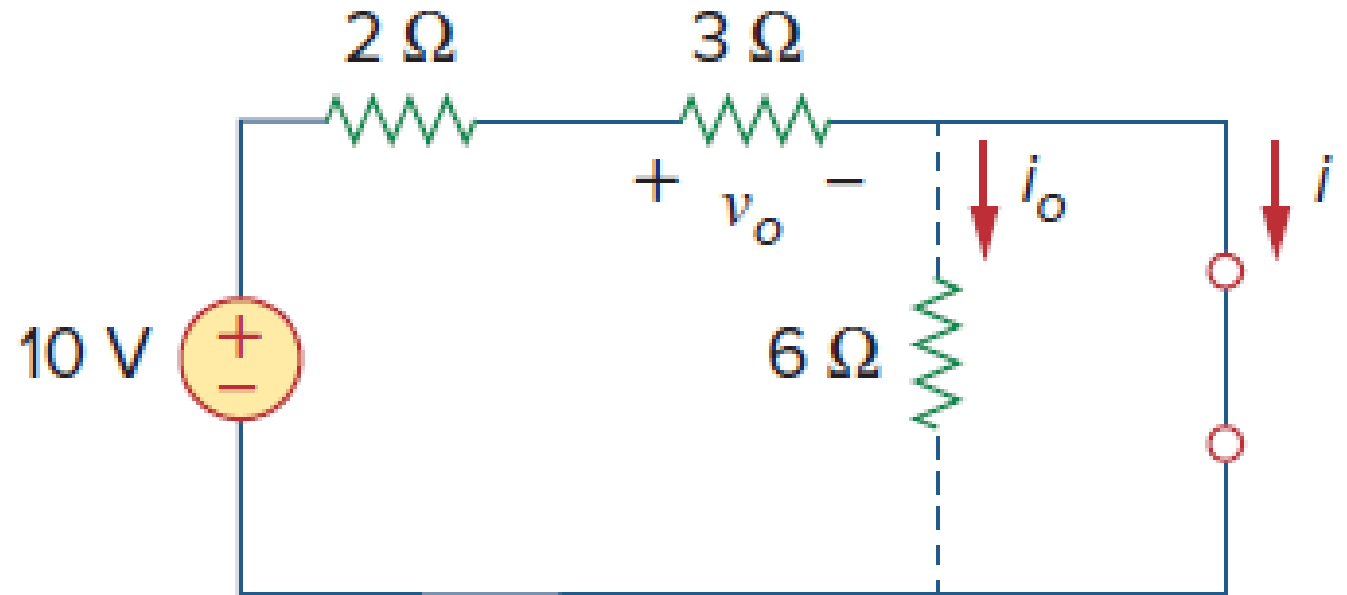
It is better to first find the inductor current i and then obtain other quantities from it.

For $t < 0$, the switch is open. Since the inductor acts like a short circuit to DC, the $6\text{-}\Omega$ resistor is short-circuited, so that we have the circuit shown

Hence, $i_o = 0$ and

$$i(t) = \frac{10}{2+3} = 2\text{ A} \quad t < 0$$

$$v_o(t) = 3i(t) = 6\text{ V} \quad t < 0$$



SOURCE-FREE RL CIRCUIT

Thus, $i(0) = 2 \text{ A}$

For $t > 0$, the switch is closed, so that the voltage source is short-circuited

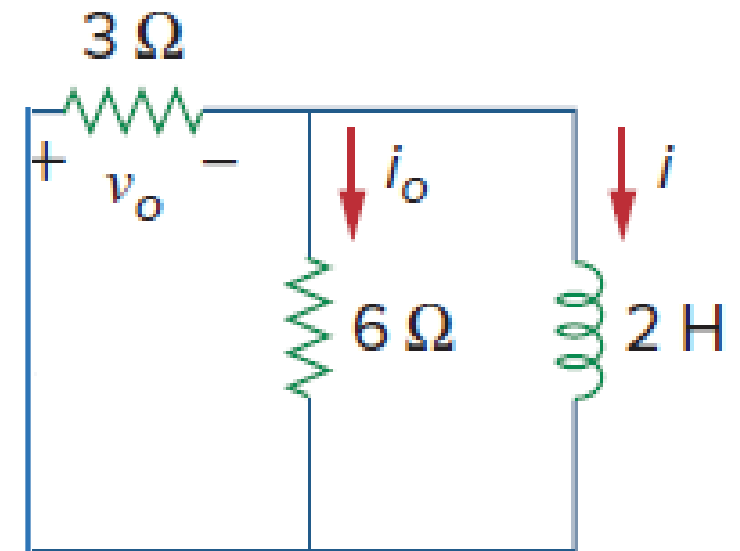
We now have a source-free RL circuit as shown. At the inductor terminals, $R_{Th} = 3 \parallel 6 = 2\Omega$

Thus the time constant is

$$\tau = \frac{L}{R_{Th}} = 1\text{s}$$

Hence,

$$i(t) = i(0)e^{-t/\tau} = 2e^{-t/\tau} \quad t > 0$$



SOURCE-FREE RL CIRCUIT

Because the inductor is in parallel with the 6- Ω and 3- Ω resistors,

$$v_o(t) = -v_L = -L \frac{di}{dt} = -2(-2e^{-t}) = 4e^{-t} \text{V}, \quad t > 0$$

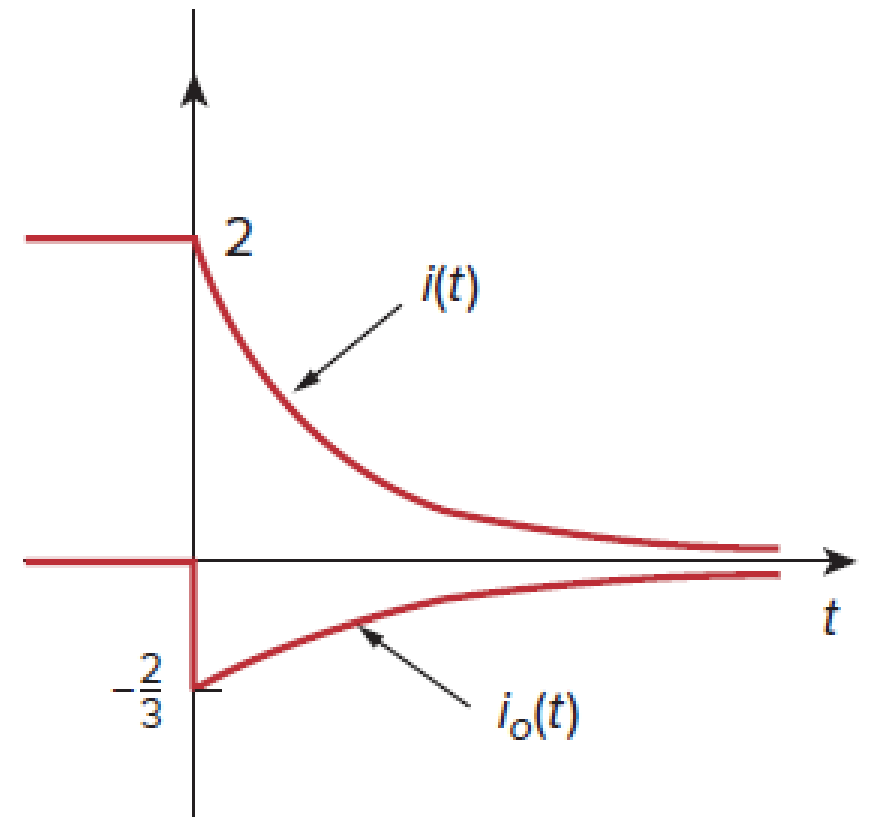
$$i_o(t) = \frac{v_L}{6} = \frac{-2}{3} e^{-t} \text{A}, \quad t > 0$$

Thus for all time,

$$i_o(t) = \begin{cases} 0 \text{A} & t < 0 \\ -\frac{2}{3} e^{-t}, & t > 0 \end{cases}$$

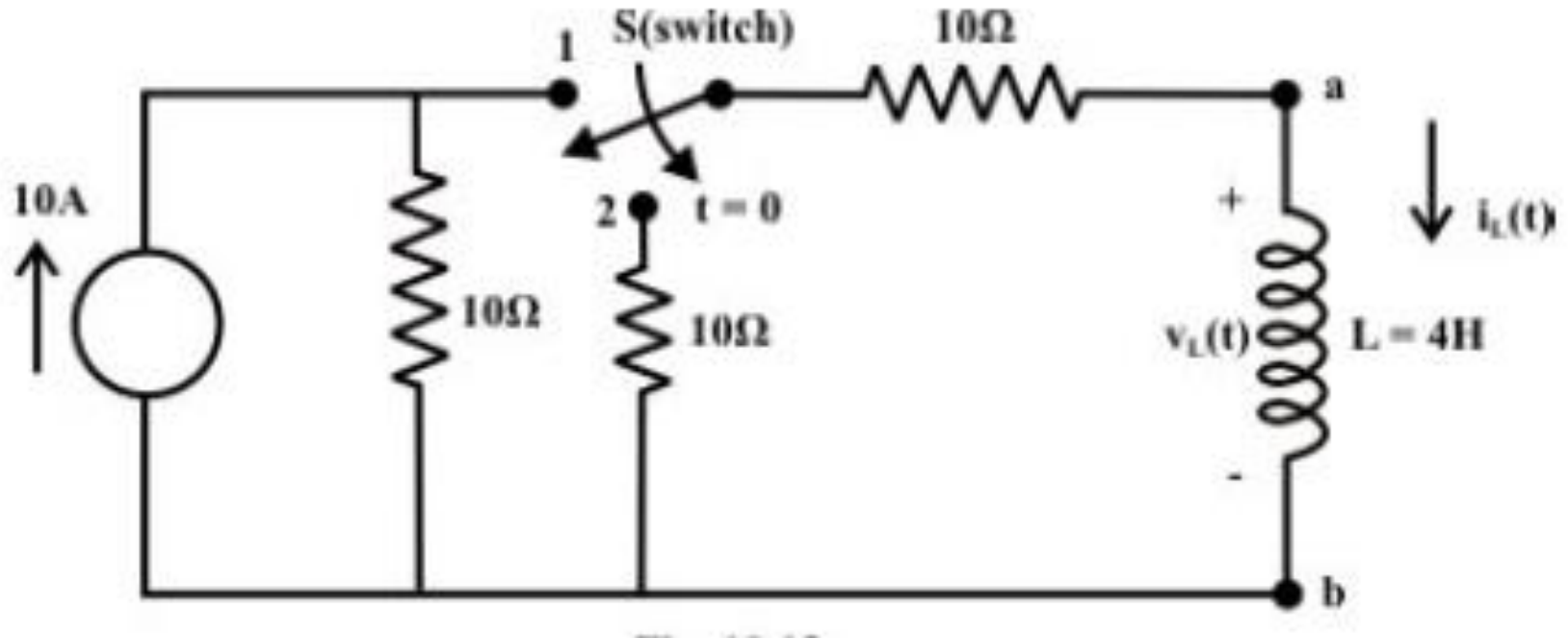
$$v_o(t) = \begin{cases} 6 \text{V} & t < 0 \\ 4e^{-t}, & t > 0 \end{cases}$$

$$i(t) = \begin{cases} 2 \text{A} & t < 0 \\ 2e^{-t}, & t \geq 0 \end{cases}$$



SOURCE-FREE RL CIRCUIT

Q6. The switch 'S' is kept in position '1' for a long time and then suddenly changed to position '2' at $t = 0$ as shown



Compute the value of v_L and i_L

- At the instant just prior to the switch changing ($t = 0^-$)
- At the instant just after the switch changes ($t = 0^+$)

Also find the rate of change of current through the inductor at $t = 0^+$

SOURCE-FREE RL CIRCUIT

Solution :

At $t = 0^-$ the current through and the voltage across the inductor are

$$i_L(0^-) = \frac{10}{10+10} \times 10 = 5A; \quad v_L(0^-) = 0V \quad (\text{inductor acts as short-circuit})$$

At $t = 0^+$, $i_L(0^+) = 5A; \quad v_L(0^+) = -(10+10) \times 5 = -100V$ (KVL)

The rate of change of current through inductor at time $t = 0^+$ is

$$L \frac{di_L(t)}{dt} \Big|_{t=0^+} = -100V \Rightarrow \frac{di_L(t)}{dt} \Big|_{t=0^+} = \frac{-100}{4} = -25A/s$$